Mahindra University
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ÉCOLE CENTRALE SCHOOL OF ENGINEERING

Mathematics-1 (MA 1101)

Tutorial Sheet-0

Topics: Limits, Continuity, Rolle's Theorem and Mean Value Theorem

1. Using the algebra (or laws) of limits, find

   (a)
   $$\lim_{x\to c} \frac{x^3+4x^2-3}{x^2+5}.$$
   In particular, evaluate the limit at $c=-1$.

   (b)
   $$\lim_{x\to -2} \sqrt{4x^2-3}.$$

2. Show that
   $$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to \infty} \frac{\sin x}{x} = 0.$$

3. Show that
   $$\lim_{x\to 0^-} x^3 \cos\left(\frac{2}{x}\right) = 0, \qquad \lim_{x\to 0^+} x^3 \cos\left(\frac{2}{x}\right) = 0.$$

4. Using
   $$\lim_{x\to 0} \frac{\sin x}{x} = 1,$$
   prove that

   (a)
   $$\lim_{x\to 0} \frac{\tan 2x}{\sin x} = 2.$$

   (b)
   $$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}.$$

   (c)
   $$\lim_{x\to 0} \frac{\sin 5x - \sin 3x}{x} = 2.$$

5. Let

$$f(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Show that f is continuous at $x = 0$.

6. Let

$$g(x) = \begin{cases} x^2 - 3, & x \leq 0, \\ \sin x, & x > 0. \end{cases}$$

Show that g is discontinuous at $x = 0$.

7. Show that on the graph of any quadratic polynomial, the chord joining the points corresponding to $x = a$ and $x = b$ is parallel to the tangent at the midpoint

$$x = \frac{a+b}{2}.$$

8. Use Rolle's theorem to prove that, regardless of the value of b, there is at most one point

$$x \in [-1, 1]$$

for which

$$x^3 - 3x + b = 0.$$

9. Show that the equation

$$x^2 = x \sin x + \cos x$$

has exactly two real solutions.

10. Show that the Mean Value Theorem can be expressed in the form

$$f(x+h) = f(x) + hf'(x+\theta h), \qquad 0 < \theta < 1.$$

Determine θ when

(a) $f(x) = x^2$.

11. Using the Mean Value Theorem, prove that

$$|\sin x - \sin y| \leq |x - y|.$$

12. Using the Mean Value Theorem, prove that

$$ny^{n-1}(x-y) \leq x^n - y^n \leq nx^{n-1}(x-y),$$

where

$$0 < y \leq x, \qquad n = 1, 2, 3, \ldots$$